OPEN ACCESS

Check for updates

Tislelizumab plus chemotherapy versus placebo plus chemotherapy as first line treatment for advanced gastric or gastro-oesophageal junction adenocarcinoma: RATIONALE-305 randomised, double blind, phase 3 trial

Miao-Zhen Qiu,[1] Do-Youn Oh,[2] Ken Kato,[3] Tobias Arkenau,[4] Josep Tabernero,[5] Marcia Cruz Correa,[6] Anastasia V Zimina,[7] Yuxian Bai,[8] Jianhua Shi,[9] Keun-Wook Lee,[10] Jufeng Wang,[11] Elena Poddubskaya,[12] Hongming Pan,[13] Sun Young Rha,[14] Ruixing Zhang,[15] Hidekazu Hirano,[16] David Spigel,[17] Kensei Yamaguchi,[18] Yee Chao,[19] Lucjan Wyrwicz,[20] Umut Disel,[21] Roberto Pazo Cid,[22] Lorenzo Fornaro,[23] Ludovic Evesque,[24] Hongwei Wang,[25] Yaling Xu,[26] Jiang Li,[25] Tao Sheng,[25] Silu Yang,[27] Liyun Li,[27] Markus Moehler,[28] Rui-Hua Xu[1]; on behalf of the RATIONALE-305 Investigators

For numbered affiliations see end of the article

Correspondence to: R-H Xu
xurh@sysucc.org.cn
(ORCID 0000-0001-9771-8534)

Additional material is published online only. To view please visit the journal online.

Cite this as: *BMJ* 2024;385:e078876
http://dx.doi.org/10.1136/bmj-2023-078876

Accepted: 11 April 2024

ABSTRACT

OBJECTIVE
To evaluate the efficacy and safety of tislelizumab added to chemotherapy as first line (primary) treatment for advanced gastric or gastro-oesophageal junction adenocarcinoma compared with placebo plus chemotherapy.

DESIGN
Randomised, double blind, placebo controlled, phase 3 study.

SETTING
146 medical centres across Asia, Europe, and North America, between 13 December 2018 and 28 February 2023.

PARTICIPANTS
1657 patients aged ≥18 years with human epidermal growth factor receptor 2 negative locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma, regardless of programmed death-ligand 1 (PD-L1) expression status, who had not received systemic anticancer therapy for advanced disease.

INTERVENTIONS
Patients were randomly (1:1) assigned to receive either tislelizumab 200 mg or placebo intravenously every three weeks in combination with chemotherapy (investigator's choice of oxaliplatin and capecitabine, or cisplatin and 5-fluorouracil) and stratified by region, PD-L1 expression, presence or absence of peritoneal metastases, and investigator's choice of chemotherapy. Treatment continued until disease progression or unacceptable toxicity.

MAIN OUTCOME MEASURES
The primary endpoint was overall survival, both in patients with a PD-L1 tumour area positivity (TAP) score of ≥5% and in all randomised patients. Safety was assessed in all those who received at least one dose of study treatment.

RESULTS
Of 1657 patients screened between 13 December 2018 and 9 February 2021, 660 were ineligible due to not meeting the eligibility criteria, withdrawal of consent, adverse events, or other reasons. Overall, 997 were randomly assigned to receive tislelizumab plus chemotherapy (n=501) or placebo plus chemotherapy (n=496). Tislelizumab plus chemotherapy showed statistically significant improvements in overall survival versus placebo plus chemotherapy in patients with a PD-L1 TAP score of ≥5% (median 17.2 months *v* 12.6 months; hazard ratio 0.74 (95% confidence interval 0.59 to 0.94); P=0.006 (interim analysis)) and in all randomised patients (median 15.0 months *v* 12.9 months; hazard ratio 0.80 (0.70 to 0.92); P=0.001 (final analysis)). Grade 3 or worse treatment related adverse events were observed in 54% (268/498) of patients in the tislelizumab plus chemotherapy arm versus 50% (246/494) in the placebo plus chemotherapy arm.

CONCLUSIONS
Tislelizumab added to chemotherapy as primary treatment for advanced or metastatic gastric or gastro-oesophageal junction adenocarcinoma provided superior overall survival with a manageable safety profile versus placebo plus chemotherapy in patients with a PD-L1 TAP score of ≥5%, and in all randomised patients.

TRIAL REGISTRATION
ClinicalTrials.gov NCT03777657

WHAT IS ALREADY KNOWN ON THIS TOPIC

Survival outcomes with platinum plus fluoropyrimidine chemotherapy alone as primary treatment for advanced gastric or gastro-oesophageal junction adenocarcinoma are poor

Results from studies with anti-programmed cell death protein 1 (PD-1) antibodies added to chemotherapy in this setting show inconsistent overall survival benefits

Debate is ongoing about the survival benefit of anti-PD-1 therapy in patients with gastric or gastro-oesophageal junction adenocarcinoma, and its magnitude according to PD-L1 expression status

WHAT THIS STUDY ADDS

Addition of the anti-PD-1 antibody tislelizumab to chemotherapy provided significant and clinically meaningful overall survival benefit versus placebo plus chemotherapy in patients with previously untreated HER2 negative advanced gastric or gastro-oesophageal junction adenocarcinoma

This finding was further enriched in patients with a PD-L1 tumour area positivity score of ≥5%

Tislelizumab plus chemotherapy could present a new primary treatment option for patients with advanced gastric or gastro-oesophageal junction adenocarcinoma

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

Introduction

Gastric cancer is one of the most common types of cancer (>1 million new diagnoses in 2020) and one of the leading causes of cancer related deaths globally (>768 000 deaths in 2020).[1] Gastric cancer is more common in Asia (~75% of new diagnoses worldwide)[2] than in Europe and North America (~16% of new diagnoses),[1 3 4] and is of particular concern in China, South Korea, and Japan.[5-7]

Before the introduction of immunotherapy, the standard of care as first line (primary) treatment for advanced human epidermal growth factor receptor 2 (HER2) negative (ie, without HER2 overexpression or amplification) gastric or gastro-oesophageal junction adenocarcinoma was platinum plus fluoropyrimidine chemotherapy,[8-10] with median overall survival rarely exceeding 12 months.[8 11] Immune checkpoint inhibitors, such as antiprogrammed cell death protein 1/programmed death-ligand 1 (PD-1/PD-L1) antibodies, have led to a paradigm shift in the standard of care of many types of solid tumours. Although several countries have approved the use of anti-PD-1 antibodies, such as nivolumab, pembrolizumab, and sintilimab, plus chemotherapy for this indication, the results from global and Asian phase 3 clinical trials in advanced gastric or gastro-oesophageal junction adenocarcinoma with these inhibitors have been inconsistent,[12-17] prompting a debate around the survival benefit of anti-PD-1 antibodies in gastric or gastro-oesophageal junction adenocarcinoma and its magnitude according to PD-L1 expression status.

Tislelizumab is an anti-PD-1 monoclonal antibody with high affinity for PD-1[18] and was specifically engineered to minimise Fcγ receptor binding on macrophages.[19 20] In a phase 2 study in Chinese patients, tislelizumab showed promising antitumour activity when added to chemotherapy as treatment for advanced gastric or gastro-oesophageal junction adenocarcinoma (objective response rate 47% and median duration of response not reached, with a median study follow-up of 15.4 months) and had an acceptable safety and tolerability profile,[21] supporting further investigation in a phase 3 study in this setting. Here, we report the primary results of the RATIONALE-305 study, which evaluated the efficacy and safety of tislelizumab plus chemotherapy compared with placebo plus chemotherapy as primary treatment for locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma.

Methods

Study design and participants

RATIONALE-305 was a randomised, double blind, global, phase 3 clinical trial conducted at 146 medical centres across Asia, Europe, and North America (see supplementary appendix). Patients aged 18 years or older with histologically confirmed, locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma and no previous systemic therapy for advanced disease were recruited. If patients had received previous neoadjuvant or adjuvant therapy, a disease progression-free interval of at least six months was required. Patients were also required to have an Eastern Cooperative Oncology Group performance status of 0 or 1 and at least one measurable or non-measurable lesion according to the Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1, as determined by investigator assessment. Patients were enrolled regardless of PD-L1 expression status. Key exclusion criteria were HER2 positive tumours, active leptomeningeal disease or uncontrolled brain metastasis, active autoimmune diseases, medical conditions requiring systemic corticosteroids or immunosuppressants, or previous treatment with anti-PD-1/PD-L1 therapies or other checkpoint inhibitors. The supplementary appendix provides a list of the eligibility criteria.

An independent data monitoring committee monitored safety and oversaw the unblinded results of a protocol specified interim overall survival analysis. The supplementary appendix provides the full protocol.

Randomisation and masking

Patients were randomly assigned (1:1) using an interactive response technology system with permuted block randomisation (block size of four) to either tislelizumab plus investigator chosen chemotherapy or placebo plus investigator chosen chemotherapy. Randomisation was stratified according to region (China (including Taiwan) *v* Japan and South Korea *v* Europe/North America), PD-L1 expression (PD-L1 tumour area positivity (TAP) score ≥5% or <5%), peritoneal metastases (yes *v* no), and investigator's choice of chemotherapy (capecitabine and oxaliplatin, or 5-fluorouracil and cisplatin). Investigators, participants, site staff, and funder staff were masked to group assignment.

Procedures

Patients were given a 200 mg fixed dose of tislelizumab or matching placebo intravenously every three weeks, in combination with investigator's choice of chemotherapy every three weeks: capecitabine 1000 mg/m^2 twice daily on days 1-14 and oxaliplatin 130 mg/m^2 on day 1 or 5-fluorouracil 800 mg/m^2 on days 1-5 and cisplatin 80 mg/m^2 on day 1 for up to six cycles. Thereafter, patients continued treatment with either tislelizumab or placebo, with optional maintenance capecitabine (only permitted for patients who initially received capecitabine and oxaliplatin), until disease progression or unacceptable toxicity. After two years of study treatment, if complete response, partial response, or stable disease were achieved, tislelizumab or placebo treatment could be stopped based on the investigator's evaluation of a patient's clinical benefit and risk. Crossover between the treatment arms or between chemotherapy regimens was prohibited.

Investigators conducted assessments of radiological tumour response by computed tomography or magnetic resonance imaging per RECIST version 1.1 about every six weeks during the first 48 weeks of the study and every nine weeks thereafter. Adverse events were

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

assessed throughout the study period and up to 30 days after the last dose of study treatment (including chemotherapy) or before initiation of new anticancer therapy, whichever occurred first. Safety was assessed by monitoring treatment emergent adverse events and serious adverse events, mapped to terms from the Medical Dictionary for Regulatory Activities version 24.0 and graded according to National Cancer Institute Common Terminology Criteria for Adverse Events (version 5.0). Treatment related adverse events included adverse events that were considered by investigators to be related to any study drug. Immune related adverse events were reported until 90 days after the last dose of tislelizumab or placebo, regardless of initiation of new anticancer therapy.

A central laboratory prospectively assessed PD-L1 expression using the TAP score, defined as total percentage of tumour area (tumour and any desmoplastic stroma) covered by tumour cells with PD-L1 membrane staining (any intensity), and tumour associated immune cells with PD-L1 staining (any intensity), visually estimated by pathologists using an investigational use only version of the Ventana PD-L1 (SP263) assay (Roche Diagnostics).[22] In addition, for exploratory purposes, pathologists in the central laboratory scored the same stained samples according to combined positive score (defined as the number of PD-L1-expressing tumour cells, lymphocytes, and macrophages divided by total number of viable tumour cells, multiplied by 100).

Outcomes

The primary endpoint was overall survival, defined as the time from randomisation to death due to any cause, assessed in patients with a PD-L1 TAP score of ≥5% and in all randomised patients (intention-to-treat population). Secondary endpoints (assessed by investigators) were progression-free survival, confirmed objective response rate, disease control rate, clinical benefit rate, time to response, and duration of response, evaluated in patients with a PD-L1 TAP score of ≥5% and in all randomised patients; safety and tolerability, assessed based on the incidence and severity of adverse events; and health related quality of life, which will be reported in a separate publication. The supplementary appendix and protocol provide a complete list of study endpoints and their definitions, together with a summary of any changes during study conduct, and the corresponding rationale.

Statistical analysis

Efficacy analyses were conducted in randomised patients with a PD-L1 TAP score of ≥5% and in all randomised patients. For the primary endpoint, a hierarchical, sequential testing method was used to control type I error at 0.025 (one sided). If the null hypothesis in the overall survival analysis in patients with a PD-L1 TAP score of ≥5% was rejected, only then was overall survival tested in all randomised patients. After a rejection of the null hypothesis for the primary endpoint, testing was shifted sequentially to secondary endpoints of progression-free survival and objective response rate in patients with a PD-L1 TAP score of ≥5%, and then in all randomised patients (see supplementary appendix for description of statistical considerations related to hierarchical testing). The inferential test was to be stopped at the first non-significant endpoint.

Assumed hazard ratios for overall survival were 0.75 in patients with a PD-L1 TAP score of ≥5%, corresponding to a median overall survival of 15.3 months with tislelizumab plus chemotherapy versus 11.5 months with placebo plus chemotherapy, and 0.80 in all randomised patients, corresponding to a median overall survival of 14.4 months versus 11.5 months, respectively. It was expected that about 384 deaths in patients with a PD-L1 TAP score of ≥5% and 768 deaths in all randomised patients at final analysis (estimated to be 48 months after the first patient was randomised) would provide a power of 80% and 87%, respectively, for superiority testing at a one sided significance level of 0.025. One protocol-specified interim analysis of overall survival was planned to take place after 70% of the target number of deaths in each population. We predefined the superiority boundary at the interim overall survival analysis using the O'Brien-Fleming boundary approximated using the Hwang-Shih-DeCani spending function. The independent data monitoring committee confirmed that at the interim analysis (data cut-off: 8 October 2021) the study met the primary endpoint of overall survival in patients with a PD-L1 TAP score of ≥5% (one sided P value boundary for superiority is 0.0092 based on 291 deaths) but not in all randomised patients (one sided P value boundary for superiority is 0.0081 based on 559 deaths), and the study continued in a blinded manner towards the final analysis. The one sided P value boundary for superiority of overall survival in all randomised patients at final analysis (data cut-off: 28 February 2023) was updated to 0.0226 based on 776 actual observed deaths.

The overall survival and progression-free survival analyses were performed using a stratified log-rank test, stratified by region (Asia *v* Europe/North America), PD-L1 expression (only for analyses in all randomised patients, PD-L1 TAP score <5% *v* ≥5%), and presence of peritoneal metastasis (yes *v* no). The stratified overall survival and progression-free survival hazard ratios and associated two sided 95% confidence intervals were estimated using a Cox proportional hazard regression model, including treatment arm as a covariate, and using stratification factors region, PD-L1 expression, and presence of peritoneal metastasis as strata. The median and cumulative probabilities of time-to-event endpoints were estimated using the Kaplan-Meier method. Overall survival was assessed in prespecified subgroups by PD-L1 expression status using TAP score, region, and several other baseline personal and disease characteristics (see supplementary appendix), as well as in post hoc subgroups by PD-L1 expression status using a combined positive score.

We calculated and compared objective response rates along with the Clopper-Pearson two sided 95%

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

confidence intervals between treatment groups. Odds ratios were calculated using the Cochran-Mantel-Haenszel method, stratified by TAP score, region, and several other baseline personal and disease characteristics (see supplementary appendix). Safety was assessed in all patients who received at least one dose of study treatment (safety population). Safety data were analysed descriptively.

More detailed statistical considerations are provided in the statistical analysis plan (see supplementary appendix) and supplementary statistical considerations (see supplementary appendix). All calculations and analyses were done using SAS (version 9.2 or higher).

Patient and public involvement

Study participants were aware of the purpose of the trial during recruitment, although they were not involved in formulating the research question, outcome measures, or study design. Most of the participants were recruited from study countries where patient and public involvement in study design was uncommon at the time the study was planned, and for those participants who were recruited in study countries where patient and public involvement was more widely practised, the sponsor was still in the process of establishing relationships with suitable patient advocacy groups. However, the results were communicated to patients who expressed an interest during visits to participating clinics.

Results

Patients

Overall, 1657 patients were screened between 13 December 2018 and 9 February 2021, of whom 997 were randomly assigned to receive either tislelizumab plus chemotherapy (n=501) or placebo plus chemotherapy (n=496; fig 1). Supplementary appendix

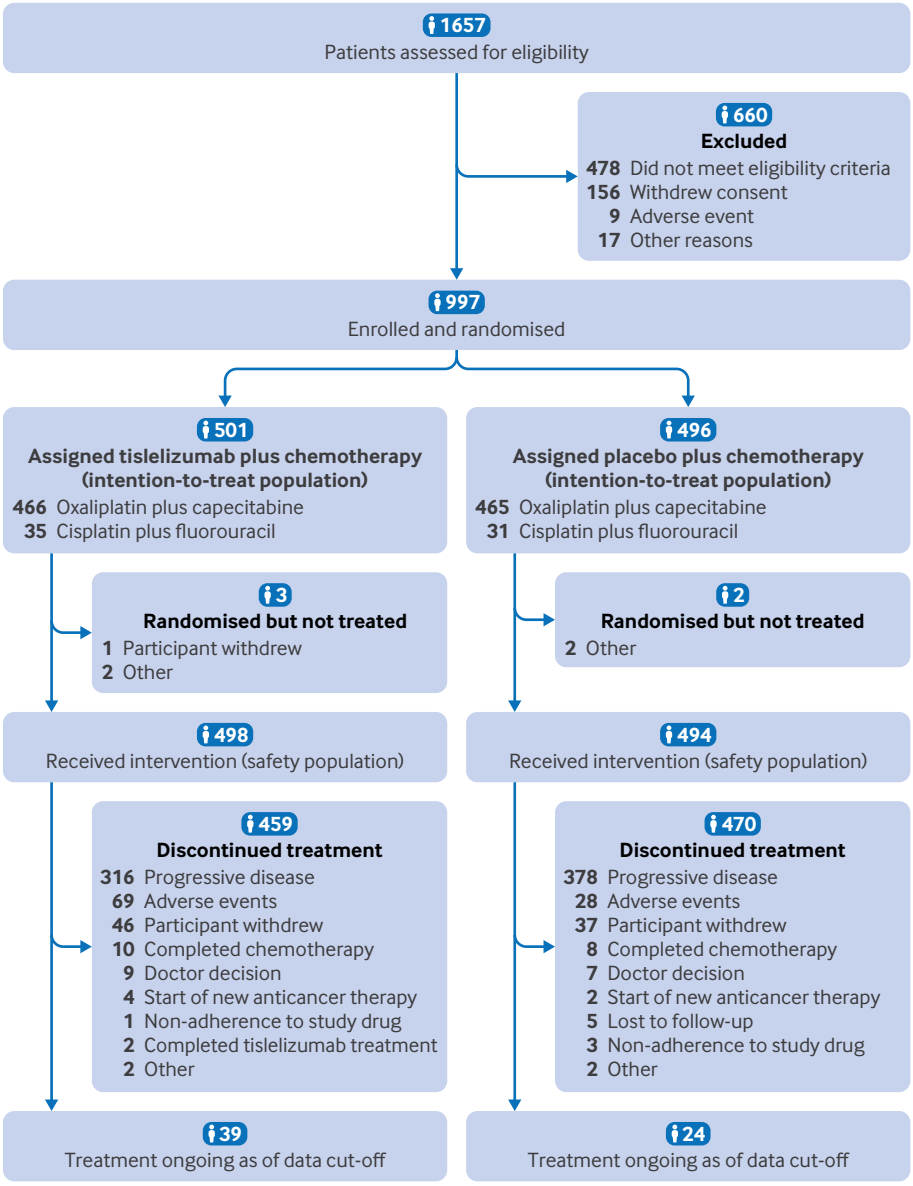

Fig 1 | Trial profile at final analysis. Data cut-off was 28 February 2023

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

table S1 presents the primary reasons for screening failure. Minimum study follow-up time (defined as time from the date of the last patient randomised to the data cut-off) was 7.9 months at the interim analysis (data cut-off: 8 October 2021) and 24.6 months at the final analysis (data cut-off: 28 February 2023). In total, 498 (99%) of 501 patients in the tislelizumab plus chemotherapy arm and 494 (>99%) of 496 in the placebo plus chemotherapy arm received at least one dose of the study drug. Personal and baseline characteristics were balanced between treatment arms (table 1). Eight hundred (80%) of 997 patients had gastric cancer and 196 (20%) had gastro-oesophageal junction adenocarcinoma. Five hundred and forty six (55%) of 997 patients had a PD-L1 TAP score of ≥5%.

Median study follow-up duration (defined as time from randomisation to data cut-off, death, or study discontinuation owing to other reasons, whichever came first for all patients) was 11.8 months (interquartile range (IQR) 7.4-15.9 months) at the interim analysis and 13.2 (IQR 7.1-24.6) months at the final analysis. At the final analysis, 459 (92%) of 501 patients in the tislelizumab plus chemotherapy arm and 470 (95%) of 496 in the placebo plus chemotherapy arm had discontinued treatment (fig 1). The median duration of exposure to chemotherapy was 5.9 (IQR 3.3-11.9) months for tislelizumab plus chemotherapy and 5.7 (IQR 3.0-9.8) months for placebo plus chemotherapy; the median durations of exposure were similar between the two arms (see supplementary appendix table S2). Overall, 276 (61%) of 453 patients in the tislelizumab plus chemotherapy arm and 250 (56%) of 450 patients in the placebo plus chemotherapy arm received capecitabine maintenance treatment (safety population). After study treatment discontinuation, 265 (53%) of 501 patients in the tislelizumab plus chemotherapy arm and 294 (59%) of 496 patients in the placebo plus chemotherapy arm received subsequent systemic anticancer therapies (see supplementary appendix table S3).

Table 1 | Baseline personal and clinical characteristics of randomised patients. Values are number (percentage) unless stated otherwise

Characteristics	Tislelizumab plus chemotherapy (n=501)	Placebo plus chemotherapy (n=496)
Median (IQR) age (years)	60.0 (53.0-66.0)	61.0 (54.0-68.0)
Sex:		
Male	346 (69)	346 (70)
Female	155 (31)	150 (30)
Race/ethnicity:		
Asian	376 (75)	372 (75)
White	116 (23)	107 (22)
Other*	9 (2)	17 (3)
Geographical region:		
Asia	376 (75)	372 (75)
North America/Europe	125 (25)	124 (25)
ECOG performance status:		
0	169 (34)	154 (31)
1	332 (66)	342 (69)
Primary tumour location:		
Stomach	405 (81)	395 (80)
Gastro-oesophageal junction	96 (19)	100 (20)†
Metastatic disease	494 (99)	490 (99)
No of metastatic sites:		
0-2	335 (67)	335 (68)
≥3	166 (33)	160 (32)
Missing	0 (0)	1 (<1)¶
Liver metastases	190 (38)	188 (38)
Peritoneal metastases	220 (44)	214 (43)
Previous adjuvant/neoadjuvant treatment	107 (21)	100 (20)
Previous gastrectomy/oesophagectomy	133 (27)	139 (28)
MSI or MMR status:		
MSI-H/dMMR	16 (3)	24 (5)
MSI-L/MSS/pMMR	448 (89)	439 (89)
Unknown	37 (7)	33 (7)
PD-L1 expression TAP score:		
<5%	227 (45)	224 (45)
≥5%	274 (55)	272 (55)
Investigator chosen chemotherapy:		
Oxaliplatin and capecitabine	466 (93)	465 (94)
Cisplatin and 5-fluouracil	35 (7)	31 (6)

Data cut-off was 28 February 2023.
dMMR=mismatch repair-deficient; ECOG=Eastern Cooperative Oncology Group; IQR=interquartile range; MSI-H/L=microsatellite instability-high/low; MSS=microsatellite stable; PD-L1=programmed death-ligand 1; pMMR=mismatch repair-proficient; TAP=tumour area positivity.
*Includes not reported, unknown, and other.
†The diagnosis of one patient was updated from gastric adenocarcinoma to pancreatic cancer after randomisation.
¶Metastatic site was surgically removed before study entry.

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

Efficacy in population with PD-L1 TAP score ≥5%

At the interim analysis, tislelizumab plus chemotherapy met the criteria for superiority versus placebo plus chemotherapy for overall survival in patients with a PD-L1 TAP score of ≥5% (median 17.2 months (95% confidence interval (CI) 13.9 to 21.3) *v* 12.6 months (12.0 to 14.4), respectively; stratified hazard ratio 0.74 (95% CI 0.59 to 0.94); P=0.006) (fig 2). Significant improvement in investigator assessed progression-free survival was observed in the tislelizumab plus chemotherapy arm versus placebo plus chemotherapy arm in patients with a PD-L1 TAP score of ≥5% (stratified hazard ratio 0.67 (0.55 to 0.83); P<0.001) (fig 3) along with higher objective response rate (non-significant) and longer duration of response (see supplementary appendix table S4 and fig S1). After an additional 17 months of follow-up (minimum 24.6 months) at the final analysis, the updated results in the tislelizumab plus chemotherapy arm versus placebo plus chemotherapy arm for overall survival (median 16.4 months (95% CI 13.6 to 19.1) *v* 12.8 months (12.0 to 14.5), respectively; stratified hazard ratio 0.71 (95% CI 0.58 to 0.86)) and for progression-free survival (0.68 (0.56 to 0.83)) in patients with a PD-L1 TAP score of ≥5% were consistent with the primary results at the interim analysis (see supplementary appendix fig S2, fig S3, and table S5).

Efficacy in intention-to-treat population

At the final analysis, tislelizumab plus chemotherapy also met the criteria for superiority versus placebo plus chemotherapy for overall survival (median 15.0 months (95% CI 13.6 to 16.5) *v* 12.9 months (12.1 to 14.1), respectively; stratified hazard ratio 0.80 (95% CI 0.70 to 0.92); P=0.001) in all randomised patients (fig 2). In the tislelizumab plus chemotherapy arm versus placebo plus chemotherapy arm, the overall survival rate at 18 months was 42% *v* 33%, respectively, and at 24 months was 33% *v* 23%, respectively (fig 2). Investigator assessed progression-free survival was also improved in the tislelizumab plus chemotherapy arm versus placebo plus chemotherapy arm in all randomised patients (stratified hazard ratio 0.78 (0.67 to 0.90)) (fig 3), along with tumour response and duration of response at the final analysis (see supplementary appendix tables S4 and S5 and fig S4).

Subgroup and exploratory analyses

Overall survival results at the final analysis consistently favoured patients in the tislelizumab plus chemotherapy arm versus the placebo plus chemotherapy arm across multiple prespecified subgroups in patients with a PD-L1 TAP score of ≥5% (see supplementary appendix table S6 and fig S5) and in all randomised patients (see supplementary appendix table S7 and fig 4). In the prespecified analysis of patients with a PD-L1 TAP score of <5%, median overall survival in the tislelizumab plus chemotherapy arm was 14.1 months (95% CI 11.9 to 15.6) *v* 12.9 months (11.3 to 14.7) in the placebo plus chemotherapy arm, with a stratified hazard ratio of 0.92 (95% CI 0.75 to 1.13). Supplementary appendix table S5 summarises the outcomes in the population with a PD-L1 TAP score of <5%. In the post hoc analysis, similar results to those seen with TAP score based subgroups were observed in patients with a PD-L1 combined positive score of ≥5 (unstratified hazard ratio 0.72 (95% CI 0.59 to 0.88)) and <5 (0.89 (0.72 to 1.09)) (see supplementary appendix table S8). Supplementary appendix tables S9 and S10 report the prevalence by PD-L1 combined positive score status and concordance between TAP score and combined positive score from the exploratory biomarker analysis.

Safety

Treatment related adverse events were reported in 483 (97%) of 498 patients in the tislelizumab plus chemotherapy arm and 476 (96%) of 494 patients in the placebo plus chemotherapy arm (see supplementary appendix table S11). Grade 3 or worse treatment related adverse events were reported in 268 (54%) of 498 patients in the tislelizumab plus chemotherapy arm and 246 (50%) of 494 patients in the placebo plus chemotherapy arm. The most common grade 3 or grade 4 treatment related adverse events were decreased neutrophil count (59 patients (12%) in the tislelizumab plus chemotherapy arm *v* 57 (12%) in the placebo plus chemotherapy arm), decreased platelet count (56 (11%) *v* 57 (12%), respectively), neutropenia (33 (7%) *v* 34 (7%), respectively), and anaemia (25 (5%) *v* 37 (7%), respectively) (table 2). Serious treatment related adverse events occurred in 113 (23%) of 498 patients in the tislelizumab plus chemotherapy arm versus 72 (15%) of 494 patients in the placebo plus chemotherapy arm (see supplementary appendix table S11). Treatment related adverse events led to treatment discontinuation in 80 (16%) of 498 patients in the tislelizumab plus chemotherapy arm versus 40 (8%) of 494 patients in the placebo plus chemotherapy arm, and dose modification in 355 (71%) versus 354 patients (72%), respectively. Treatment related adverse events that led to death occurred in six of 498 patients (1%) in the tislelizumab plus chemotherapy arm (unspecified death (n=4), colitis (n=1), and sepsis (n=1)) and two of 494 patients (<1%) in the placebo plus chemotherapy arm (pneumonia (n=2)). A total of 154 (31%) of 498 patients in the tislelizumab plus chemotherapy arm versus 58 (12%) of 494 patients in the placebo plus chemotherapy arm experienced immune mediated adverse events, with 38 (8%) *v* 10 (2%) patients, respectively, experiencing grade 3 or worse events. Supplementary appendix table S12 lists the most common immune mediated adverse events reported.

Discussion

Primary treatment of advanced gastric or gastro-oesophageal junction adenocarcinoma with tislelizumab plus chemotherapy provided significant overall survival benefit versus placebo plus chemotherapy in patients with previously untreated HER2 negative advanced gastric or gastro-oesophageal junction adenocarcinoma. At the final analysis, overall survival rates at 18 and 24 months were higher with

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

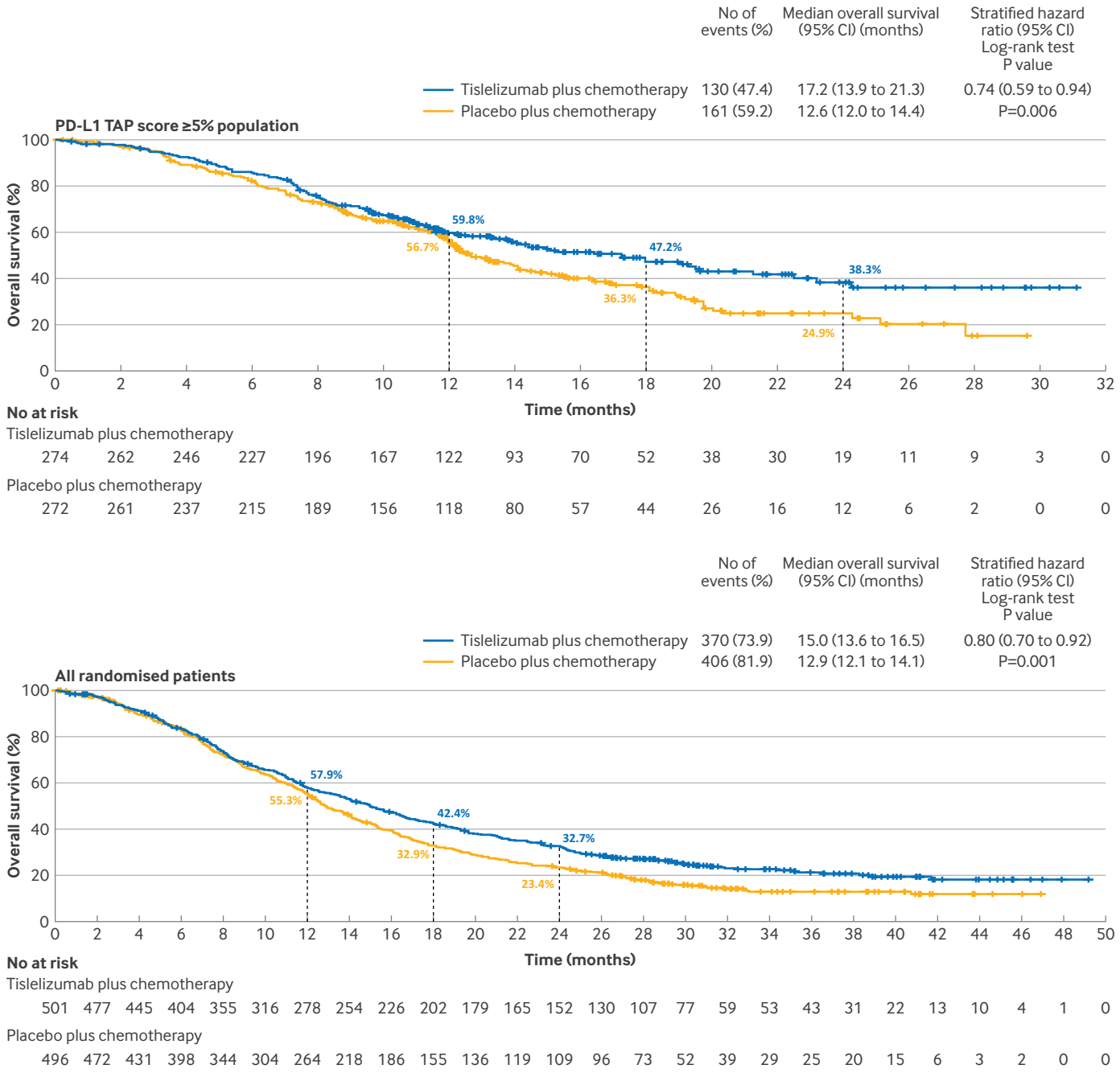

Fig 2 | Kaplan-Meier plots of overall survival in population with PD-L1 TAP scores of ≥5% (interim analysis, data cut-off: 8 October 2021) and in all randomised patients (final analysis, data cut-off: 28 February 2023). Log-rank and Cox regression models were stratified by region (Asia *v* other regions), PD-L1 expression (all randomised patients), and presence of peritoneal metastasis. P values are one sided and based on the stratified log-rank test. CI=confidence interval; PD-L1=programmed death-ligand 1; TAP=tumour area positivity

tislelizumab plus chemotherapy in patients with a PD-L1 TAP score of ≥5%, and in all randomised patients. With risk reductions for death of 29% and 20% in patients with a PD-L1 TAP score of ≥5% and in all randomised patients, respectively, the magnitude of improvement was considered clinically meaningful.[23] The overall survival benefit was observed across multiple prespecified patient subgroups and was accompanied by improvements in progression-free survival, objective response rates, and duration of response. With a minimum study follow-up time of 24.6 months and death events reported in 78% of all randomised patients at the final analysis, the current study provides robust data for efficacy and safety.

Comparison with other studies

During the conduct of the RATIONALE-305 study, other phase 3 trials of anti-PD-1 antibodies for primary

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

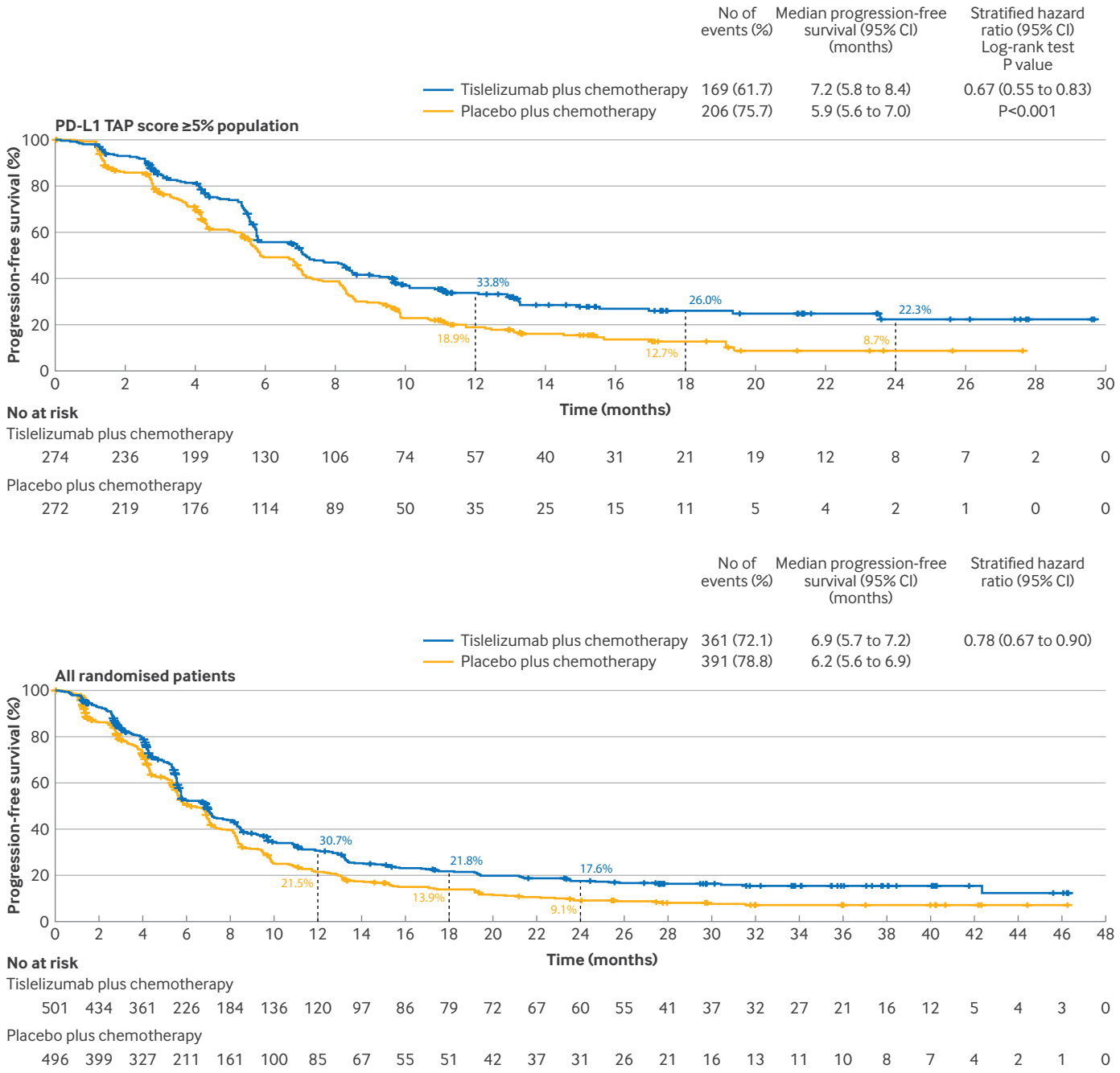

Fig 3 | Kaplan-Meier plots of progression-free survival in population with PD-L1 TAP scores of ≥5% (interim analysis, data cut-off: 8 October 2021) and all randomised patients (final analysis, data cut-off: 28 February 2023). Log-rank and Cox regression models were stratified by region (east Asia *v* other regions), PD-L1 expression (all randomised patients), and presence of peritoneal metastasis. P values are one sided and based on the stratified log-rank test. CI=confidence interval; PD-L1=programmed death-ligand 1; TAP=tumour area positivity

treatment of advanced gastric or gastro-oesophageal junction adenocarcinoma (in both global and Asian populations) have been reported, with variable observed results for overall survival. Nivolumab plus chemotherapy showed a significant overall survival benefit versus chemotherapy alone in patients with a PD-L1 combined positive score of ≥5 (hazard ratio 0.71) and in all randomised patients (hazard ratio 0.80) in a global population in the CheckMate 649 trial,[13] but not in all randomised patients in an Asian population in the ATTRACTION-4 (nivolumab plus chemotherapy versus chemotherapy alone in patients with HER2 negative, untreated, unresectable advanced or recurrent gastric or gastro-oesophageal junction cancer) trial (hazard ratio 0.90).[14] Pembrolizumab plus chemotherapy failed to show superiority versus chemotherapy alone in patients with high expression of PD-L1 in the KEYNOTE-062 (efficacy and safety of

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

	No of events/No of patients		Unstratified hazard ratio (95% CI)
	Tislelizumab plus chemotherapy	Placebo plus chemotherapy	
Overall	370/501	406/496	0.80 (0.70 to 0.92)
Age (years)			
<65	258/340	265/313	0.81 (0.68 to 0.96)
≥65	112/161	141/183	0.79 (0.61 to 1.01)
Sex			
Male	252/346	280/346	0.81 (0.68 to 0.96)
Female	118/155	126/150	0.79 (0.61 to 1.02)
Race/ethnicity			
Asian	274/376	298/372	0.83 (0.70 to 0.97)
White	91/116	92/107	0.71 (0.53 to 0.95)
Other	5/9	16/17	0.53 (0.19 to 1.48)
Geographical region			
Asia	274/376	298/372	0.83 (0.70 to 0.97)
Europe or North America	96/125	108/124	0.71 (0.54 to 0.94)
ECOG performance status			
0	127/169	123/154	0.79 (0.62 to 1.01)
1	243/332	283/342	0.80 (0.68 to 0.96)
Primary location			
Gastro-oesophageal junction	68/96	82/100	0.70 (0.51 to 0.97)
Stomach	302/405	323/395	0.83 (0.71 to 0.97)
Disease stage at screening			
Locally recurrent or advanced	3/7	5/5	0.21 (0.04 to 1.11)
Metastatic	367/494	400/490	0.81 (0.70 to 0.94)
No of metastatic sites at study entry			
0-2	237/335	263/335	0.77 (0.65 to 0.92)
≥3	133/166	143/160	0.84 (0.66 to 1.06)
Presence of liver metastasis at study entry			
Yes	137/190	161/188	0.75 (0.60 to 0.95)
No	233/311	245/308	0.83 (0.70 to 1.00)
Presence of peritoneal metastasis at study entry			
Yes	177/220	188/214	0.80 (0.65 to 0.98)
No	193/281	218/282	0.79 (0.65 to 0.95)
Previous adjuvant or neoadjuvant therapy			
Yes	78/107	89/100	0.68 (0.50 to 0.92)
No	292/394	317/396	0.83 (0.71 to 0.98)
Previous gastrectomy or oesophagectomy			
Yes	93/133	112/139	0.78 (0.59 to 1.03)
No	277/368	294/357	0.81 (0.68 to 0.95)
MSI or MMR status			
MSI-H/dMMR	10/16	18/24	0.66 (0.30 to 1.43)
MSI-L/MSS/pMMR	335/448	362/439	0.82 (0.70 to 0.95)
Unknown	25/37	26/33	0.66 (0.38 to 1.15)
Investigator's choice of chemotherapy			
Oxaliplatin and capecitabine	340/466	379/465	0.79 (0.68 to 0.91)
Cisplatin and 5-fluorouracil	30/35	27/31	0.89 (0.53 to 1.51)

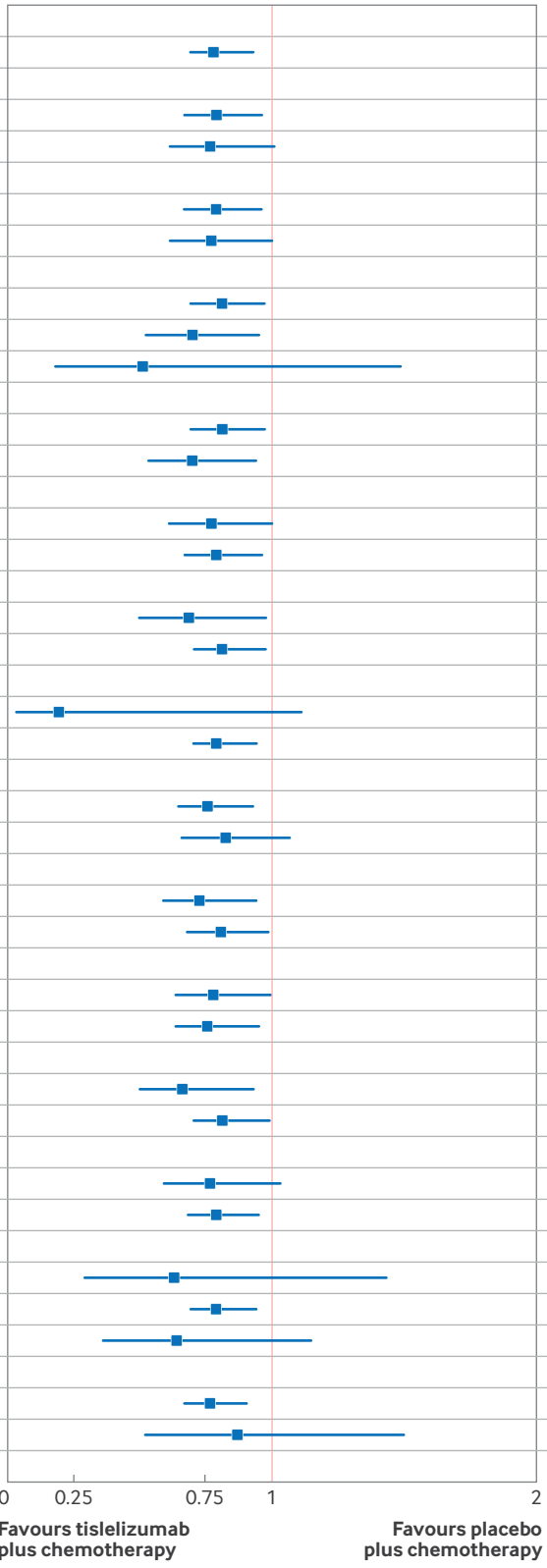

Fig 4 | Prespecified subgroup analysis of overall survival in all randomised patients (final analysis data cut-off: 28 February 2023). Medians were estimated by the Kaplan-Meier method with 95% CIs estimated using the method of Brookmeyer and Crowley using log-log transformation. Hazard ratios and corresponding 95% CIs were estimated from unstratified Cox regression model including treatment as covariate. The race subcategory "other" includes not reported, unknown, and other. CI=confidence interval; dMMR=mismatch repair-deficient; ECOG=Eastern Cooperative Oncology Group; MSI-L/H=microsatellite instability-low/high; MSS=microsatellite stable; MMR=mismatch repair-proficient

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

Table 2 | Treatment related adverse events with an incidence ≥10% by preferred term and worst grade (safety population). Values are number (percentage) of patients by worst grade of event

	Tislelizumab plus chemotherapy (n=498)				Placebo plus chemotherapy (n=494)			
Adverse events	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
Any event	215 (43)	231 (46)	26 (5)	11 (2)	230 (47)	217 (44)	25 (5)	4 (<1)
Nausea	224 (45)	13 (3)	0 (0)	0 (0)	223 (45)	9 (2)	0 (0)	0 (0)
Decreased appetite	168 (34)	14 (3)	0 (0)	0 (0)	169 (34)	16 (3)	0 (0)	0 (0)
Decreased platelet count	118 (24)	41 (8)	15 (3)	0 (0)	126 (26)	46 (9)	11 (2)	0 (0)
Decreased neutrophil count	109 (22)	54 (11)	5 (1)	0 (0)	103 (21)	53 (11)	4 (<1)	0 (0)
Vomiting	150 (30)	11 (2)	0 (0)	0 (0)	150 (30)	12 (2)	0 (0)	0 (0)
Anaemia	133 (27)	25 (5)	0 (0)	0 (0)	126 (26)	35 (7)	2 (<1)	0 (0)
Increased aspartate aminotransferase	132 (27)	12 (2)	1 (<1)	0 (0)	133 (27)	4 (<1)	0 (0)	0 (0)
Decreased white blood cell count	104 (21)	14 (3)	1 (<1)	0 (0)	126 (26)	8 (2)	0 (0)	0 (0)
Increased alanine aminotransferase	105 (21)	8 (2)	0 (0)	0 (0)	93 (19)	4 (<1)	0 (0)	0 (0)
Diarrhoea	99 (20)	12 (2)	0 (0)	0 (0)	115 (23)	11 (2)	0 (0)	0 (0)
Peripheral sensory neuropathy	105 (21)	1 (<1)	0 (0)	0 (0)	113 (23)	3 (<1)	0 (0)	0 (0)
Palmar-plantar erythrodysaesthesia syndrome	80 (16)	15 (3)	0 (0)	0 (0)	83 (17)	10 (2)	0 (0)	0 (0)
Asthenia	66 (13)	10 (2)	0 (0)	0 (0)	64 (13)	7 (1)	0 (0)	0 (0)
Fatigue	66 (13)	9 (2)	0 (0)	0 (0)	55 (11)	6 (1)	0 (0)	0 (0)
Neutropenia	41 (8)	32 (6)	1 (<1)	0 (0)	46 (9)	32 (7)	2 (<1)	0 (0)
Hypoaesthesia	68 (14)	1 (<1)	0 (0)	0 (0)	67 (14)	0 (0)	0 (0)	0 (0)
Increased blood bilirubin	54 (11)	6 (1)	1 (<1)	0 (0)	55 (11)	3 (<1)	0 (0)	0 (0)
Thrombocytopenia	45 (9)	14 (3)	1 (<1)	0 (0)	42 (9)	12 (2)	2 (<1)	0 (0)
Decreased weight	58 (12)	0 (0)	0 (0)	0 (0)	53 (11)	0 (0)	0 (0)	0 (0)
Hypothyroidism	54 (11)	1 (<1)	0 (0)	0 (0)	12 (2)	0 (0)	0 (0)	0 (0)

Data cut-off was 28 February 2023.
Data are shown for all grade incidence of ≥10% in either treatment arm (see also supplementary appendix table S11).
Treatment related adverse events are sorted by decreasing frequency for all grade events in the tislelizumab plus chemotherapy arm. Patients with two or more adverse events in the same preferred term are counted only once for that preferred term. Adverse events were graded based on National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0 and coded using Medical Dictionary for Regulatory Activities version 24.0.

pembrolizumab or pembrolizumab plus chemotherapy versus chemotherapy alone for patients with first line, advanced gastric cancer) trial (combined positive score ≥1 or combined positive score ≥10),[15] although significant overall survival benefit was shown versus chemotherapy alone in patients with a PD-L1 combined positive score of ≥10 (hazard ratio 0.65) or ≥1 (hazard ratio 0.74), and in all randomised patients (hazard ratio 0.78) in the KEYNOTE-859 (pembrolizumab plus chemotherapy versus chemotherapy alone for HER2 negative advanced gastric cancer) trial.[16] Another anti-PD-1 antibody, sintilimab, has shown significantly improved overall survival in combination with chemotherapy versus chemotherapy alone in Chinese patients with PD-L1 combined positive scores of ≥5 (hazard ratio 0.66) and in all randomised patients (hazard ratio 0.77) in the ORIENT-16 (sintilimab plus chemotherapy for unresectable gastric or gastro-oesophageal junction cancer) trial.[17] The variability in overall survival observed among these studies could potentially be attributed to differences in study design, in the proportion of Asian compared with European/North American patients of the included populations, in chemotherapy regimens, or in the studies' statistical considerations, as well as differences in use of subsequent therapies. With the inclusion of a similar study population and use of a similar chemotherapy regimen, but with a slightly higher number of patients receiving subsequent immunotherapy, the overall survival benefit of tislelizumab plus chemotherapy versus placebo plus chemotherapy in RATIONALE-305 (hazard ratios 0.74 in the population with a PD-L1 TAP score ≥5% and 0.80 in all randomised patients) was consistent with the observed benefits in the global studies, with the addition of nivolumab or pembrolizumab to chemotherapy in CheckMate 649[13] and KEYNOTE-859,[16] respectively. These data for tislelizumab reinforce the survival benefit seen with anti-PD-1 antibodies in combination with chemotherapy as primary treatment for patients with gastric or gastro-oesophageal junction adenocarcinoma.

The overall survival benefit of tislelizumab plus chemotherapy appeared greater in the population with a PD-L1 TAP score of ≥5% compared with patients with a score of <5%, suggesting an enrichment of survival benefit in patients with higher PD-L1 expression than in patients with lower PD-L1 expression. In patients with a PD-L1 TAP score of <5%, the hazard ratio for overall survival seemed to favour the tislelizumab plus chemotherapy arm over the placebo plus chemotherapy arm (ie, hazard ratio <1); a similar trend in the hazard ratio for progression-free survival and a numerical increase in objective response rate were also observed. This prespecified exploratory analysis was not powered for hypothesis testing, however, and should be interpreted with caution. The observed difference in the magnitude of survival benefit between patients with high and low PD-L1 expression was similar to findings in studies of other anti-PD-1 antibodies plus chemotherapy in the setting of primary treatment of patients with advanced gastric or gastro-oesophageal junction adenocarcinoma.[13 16 17] We acknowledge that challenges in data interpretation across studies may

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

arise when different PD-L1 scoring methods are used. The prevalence of patients with a PD-L1 TAP score of ≥5% was similar to that for a combined positive score of ≥5 in other studies in gastric or gastro-oesophageal junction adenocarcinoma.[13 17] Reassuringly, we observed an acceptable concordance rate between TAP scores of ≥5% and combined positive scores of ≥5 in the exploratory analysis in our study, and similar overall survival results with tislelizumab plus chemotherapy in subgroups defined by TAP score or combined positive score in the exploratory analysis.

The RATIONALE-305 study population was representative of the global distribution of gastric or gastro-oesophageal junction adenocarcinoma, with 75% of patients enrolled from Asia and 25% from Europe and North America; the data therefore provide valuable clinical evidence relevant to the management of patients with gastric or gastro-oesophageal junction adenocarcinoma globally. Subgroup analyses showed that overall survival benefit with tislelizumab plus chemotherapy was consistently seen across the Asia and Europe/North America regions in the population with a PD-L1 TAP score of ≥5%, and in all randomised patients. A consistent overall survival benefit among different regional subgroups was also observed with other anti-PD-1 antibodies in the CheckMate 649 and KEYNOTE-859 studies,[13 16] despite numerical differences in the magnitude of treatment effect across subgroups, which should be interpreted with caution owing to the exploratory nature of subgroup analyses. Similar to results observed with other anti-PD-1 antibodies,[13 16] survival benefit with tislelizumab plus chemotherapy was observed regardless of microsatellite instability status. Results suggested that the magnitude of benefit was greater in patients with high microsatellite instability/mismatch repair-deficient tumours than those with low microsatellite instability/mismatch repair-proficient tumours, although the wide confidence interval, likely due to the small sample size, precludes a definitive conclusion (fig 4).

The safety profile of tislelizumab plus chemotherapy was manageable when being used as primary treatment in patients with locally advanced unresectable or metastatic gastric or gastro-oesophageal junction adenocarcinoma. The duration of treatment with tislelizumab or placebo, along with chemotherapy, was similar in both arms. The most common treatment related adverse events (any grade and grade ≥3) in both groups were consistent with the known safety profile of the individual study treatment components. Most immune mediated adverse events were grade 1 or grade 2 in severity, and the profile of commonly reported immune mediated adverse events was generally similar to those seen with other anti-PD-1 antibodies.[13-17 24] Overall, the toxicity of chemotherapy did not appear to worsen with the addition of tislelizumab.

Limitation of this study

A potential limitation of the current study was the lack of independent review committee assessment of tumour responses. As the investigators and site staff were blinded to group assignment and PD-L1 expression, however, the potential for bias in investigator assessed responses using standard criteria was expected to be minimal.

Conclusions

Primary treatment of advanced gastric or gastro-oesophageal junction adenocarcinoma with tislelizumab plus chemotherapy provided significant and clinically meaningful overall survival benefit versus placebo plus chemotherapy in patients with a PD-L1 TAP score of ≥5%, and in all randomised patients. The safety profile of tislelizumab in combination with chemotherapy was manageable and acceptable. These data suggest that tislelizumab plus chemotherapy represents a potential new primary treatment option for patients with advanced gastric or gastro-oesophageal junction adenocarcinoma.

AUTHOR AFFILIATIONS

[1]Department of Medical Oncology, Sun Yat-sen University Cancer Center, State Key Laboratory of Oncology in South China, Collaborative Innovation Center of Cancer Medicine, Research Unit of Precision Diagnosis and Treatment for Gastrointestinal Cancer, Chinese Academy of Medical Sciences, Guangzhou, China

[2]Seoul National University Hospital, Cancer Research Institute, Seoul National University College of Medicine, Integrated Major in Innovative Medical Science, Seoul National University Graduate School, Seoul, South Korea

[3]Department of Head and Neck, Esophageal Medical Oncology and Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

[4]Sarah Cannon Research, London, UK

[5]Medical Oncology Department, Vall d'Hebron University Hospital, Barcelona, Spain

[6]University of Puerto Rico, San Juan, Puerto Rico

[7]BHI of Omsk Region Clinical Oncology Dispensary, Omsk, Russia

[8]Harbin Medical University Cancer Hospital, Harbin, China

[9]Linyi Cancer Hospital, Linyi, China

[10]Seoul National University College of Medicine, Seoul National University Bundang Hospital, Seongnam, South Korea

[11]Henan Cancer Hospital, Zhengzhou, China

[12]Vitamed, Moscow, Russia

[13]Oncology Department, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, Hangzhou, China

[14]Yonsei Cancer Center, Yonsei University College of Medicine, Yonsei University Health System, Seoul, South Korea

[15]Fourth Hospital of Hebei Medical University, Shijiazhuang, China

[16]Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

[17]Sarah Cannon Research Institute, Tennessee Oncology, Nashville, TN, USA

[18]Gastroenterological Chemotherapy, Cancer Institute Hospital of JFCR, Tokyo, Japan

[19]Department of Oncology, Taipei Veterans General Hospital, Taipei, China

[20]Narodowy Instytut Onkologii, Warsaw, Poland

[21]Acibadem Adana Hospital, Adana, Turkey

[22]Hospital Universitario Miguel Servet, Zaragoza, Spain

[23]Azienda Ospedaliero Universitaria Pisana, Pisa, Italy

[24]Centre Antoine Lacassagne, Nice, France

[25]BeiGene, Boston, MA, USA

[26]BeiGene, Shanghai, China

[27]BeiGene, Beijing, China

[28]Gastrointestinal Oncology, Johannes Gutenberg-University Clinic, Mainz, Germany

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

We thank Dewan Zang and Jin Wang of BeiGene for their contributions to the study design, and the study participants and study staff for their contributions to the conduct of the study. Medical writing support and editorial assistance, under the direction of the authors, was provided by Simon Lancaster and Lynda McEvoy of Ashfield MedComms, an Inizio company, and was funded by BeiGene. The list of RATIONALE-305 investigators is included in the supplementary appendix.

Contributors: M-ZQ, D-YO, and MM contributed equally to this work and are joint first authors. R-HX is the guarantor. D-YO, KK, TA, JT, MCC, MM, and R-HX were steering committee members. R-HX, LL, TS, JL, YX, and HW conceived and designed the study and acquired, analysed, and interpreted the data. MM conceived and designed the study and analysed and interpreted the data. SY conceived and designed the study and acquired and interpreted the data. M-ZQ, D-YO, KK, TA, JT, MCC, AVZ, YB, JS, K-WL, JW, EP, HP, SYR, RZ, HH, DS, KY, YC, LW, UD, RPC, LF, and LE acquired and interpreted the data. M-ZQ, D-YO, MM, and R-HX had direct access to the data and verified the underlying data reported in the manuscript. All authors participated in writing or reviewing the manuscript. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Funding: This study was funded by BeiGene. The funder had a role in study design, data collection, data analysis, data interpretation and writing of the clinical study report, and provided medical writing support. The clinical researchers were independent from the funder, and all authors had access to all relevant aggregated study data and provided final approval for the decision to submit the manuscript for publication. All authors take responsibility for the integrity of the data and the accuracy of the data analysis.

Competing interests: All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: support from BeiGene. D-YO has participated on a data safety monitoring board or advisory board for AstraZeneca, Novartis, Genentech/F Hoffmann-La Roche, Merck Serono, Bayer, Taiho, ASLAN, Halozyme, Zymeworks, Bristol Myers Squibb/Celgene, BeiGene, Basilea, Turning Point, Yuhan, Arcus Biosciences, IQVIA, and Merck Sharp and Dohme; and has received grants from AstraZeneca, Novartis, Array, Eli Lilly, Servier, BeiGene, Merck Sharp and Dohme, and Handok. KK has received consulting fees from Bristol Myers Squibb, Ono Pharmaceutical, Merck Sharp and Dohme, Bayer, BeiGene, AstraZeneca, Seagen/Pfizer, Servier, and Janssen; and has received honorariums from Bristol Myers Squibb, Ono Pharmaceutical, and Merck Sharp and Dohme. TA has participated on a data safety monitoring board or advisory board for BeiGene; and has received honorariums from iOnctura and LabGenius. JT has received consulting fees from Array Biopharma, AstraZeneca, Bayer, Boehringer Ingelheim, Cardiff Oncology, Chugai, Daiichi Sankyo, Genentech/F Hoffmann-La Roche, HalioDx, Hutchison MediPharma International, Ikena Oncology, Inspirna, IQVIA, Lilly, Menarini, Merck Serono, Merus, Merck Sharp and Dohme, Mirati, NeoPhore, Novartis, Ona Therapeutics, Orion Biotechnology, Peptomyc, Pfizer, Pierre Fabre, Samsung Bioepis, Sanofi, Scandion Oncology, Scorpion Therapeutics, Seagen/Pfizer, Servier, Sotio Biotech, Taiho Pharmaceutical, TheraMyc, and Tolremo Therapeutics; has received honorariums from Imedex/HMP, Medscape Education, MJH Life Sciences, PeerView Institute for Medical Education, and Physicians Education Resource; and has stock in Oniria Therapeutics. MCC has received grants from BeiGene, AbbVie, Genentech/F Hoffmann-La Roche, Taiho, Seagen/Pfizer, Bristol Myers Squibb, Merck, Pfizer, Janssen, Mirati Therapeutics, Tempus, HUYABIO International, Regeneron, and DELFI. K-WL has received support for the present manuscript from BeiGene; has received grants to his institution for conducting clinical trials from AstraZeneca, Ono Pharmaceutical, Merck Sharp and Dohme, Merck KGaA, F Hoffmann-La Roche, Pfizer, Leap Therapeutics, ALX Oncology, Zymeworks, Astellas, Macrogenics, Amgen, Seagen/Pfizer, Bolt Therapeutics, Trishula Therapeutics, Oncologie, Pharmacyclics, MedPacto, Green Cross, ABL Bio, Y-BIOLOGICS, Daiichi Sankyo, Taiho Pharmaceutical, InventisBio, Elevar Therapeutics, Metafines, Idience, Genome and Company, and Exelixis; has received honorariums from Ono Pharmaceutical, Boryung, Daiichi Sankyo, Astellas, and Sanofi-Aventis; and has participated on a data safety monitoring board or advisory board for ALX Oncology and Metafines. SYR has received grants from AstraZeneca, Ono Pharmaceutical, Eisai, Ipsen, Merck Sharp and Dohme, Merck KGaA, Pfizer, BeiGene, Astellas Pharma, Amgen, ALX Oncology, Zymeworks, Macrogenics, Seagen/Pfizer, Bold Therapeutics, MedPacto, ABLBIO, Daiichi Sankyo, Taiho Pharmaceutical, Leap Therapeutics, and Arcus Biosciences; has received consulting fees from LG Biochem and Indivumed; has received honorariums from Merck Sharp and Dohme, Lilly, Daiichi Sankyo, and Eisai; and has participated on a data safety monitoring board or advisory board for Amgen. HH has received support for the present manuscript from BeiGene; has received grants from Amgen, Astellas, Bristol Myers Squibb, Chugai Pharmaceutical, Daiichi Sankyo, Eisai, Insyte, Janssen Pharmaceutical, Merck Biopharma, Merck Sharp and Dohme, Novartis, Ono Pharmaceutical, Pfizer, Seagen/Pfizer, and Taiho Pharmaceutical; and has received honorariums from Nichi-Iko, Novartis, Ono Pharmaceutical, Taiho Pharmaceutical, and Teijin Pharma. DS has received support for the present manuscript from BeiGene; has received grants from Genentech/F Hoffmann-La Roche, Novartis, Celgene, Bristol Myers Squibb, Lilly, AstraZeneca, University of Texas of SW Medical Center - Simmons Cancer Center, Merck, G1 Therapeutics, Neon Therapeutics, Nektar, Celldex, Clovis Oncology, Daiichi Sankyo, Astellas Pharma, GRAIL, Transgene, Aeglea Biotherapeutics, Ipsen, BIND Therapeutics, Eisai, ImClone Systems, Janssen Oncology, MedImmune, Agios, GlaxoSmithKline, Tesaro, Cyteir Therapeutics, Novocure, Elevation Oncology, Calithera Biosciences, Arcus Biosciences, Arrys Therapeutics, Bayer, BeiGene, Blueprint Medicine, Boehringer Ingelheim, Hutchinson MediPharma, Incyte, Kronos Bio, Loxo Oncology, MacroGenics, Molecular Templates, Pure Tech Health, Razor Genomics, Repare Therapeutics, Rgenix, Tizona Therapeutics, Verastem, BioNTech, AbbVie, Amgen, Anheart Therapeutics, Ascendis Pharma, Endeavour BioMedicines, Erasca, Faeth Therapeutics, FujiFilm, Gilead Sciences, Jazz Pharmaceuticals, Lyell Immunopharma, Millennium, Moderna Therapeutics, Monte Rosa Therapeutics, Peloton Therapeutics, Shenzhen Chipscreen Biosciences, Stemline Therapeutics, Synthekine, Taiho Oncology, Tango Therapeutics, Tarveda Therapeutics, Zai Laboratory, Apollomics, Strata Oncology, and Asher Biotherapeutics; and has received consulting fees from Genentech/F Hoffmann-La Roche, Novartis, Bristol Myers Squibb, AstraZeneca, GlaxoSmithKline, Molecular Templates, Jazz Pharmaceuticals, Sanofi-Aventis, Regeneron, Lilly, BeiGene, Ipsen, Monte Rosa Therapeutics, AbbVie, Lyell Immunopharma, and Novocure. KY has received support for the present manuscript from BeiGene; has received grants from Taiho Pharmaceutical; and has received honorariums from Daiichi Sankyo, Chugai Pharmaceutical, Bristol Myers Squibb K.K., Eli Lilly Japan K.K., Taiho Pharmaceutical, Ono Pharmaceutical, Takeda Pharmaceutical, and Merck Biopharm. RPC has received grants from Astellas and Ipsen; has received honorariums from Eisai, F Hoffmann-La Roche, and Bristol Myers Squibb; has received payment for expert testimony from AstraZeneca and Lilly; and has received support for attending meetings and/or travel from Lilly, F Hoffmann-La Roche and Bristol Myers Squibb. LF has received grants from AstraZeneca; has received consulting fees from AstraZeneca, Merck Sharp and Dohme, Eli Lilly, and Servier; has received honorariums from Merck Sharp and Dohme, Eli Lilly, and Bristol Myers Squibb; and has participated on a data safety monitoring board or advisory board for Merck Sharp and Dohme, Bristol Myers Squibb, Daiichi Sankyo, Servier, Taiho Oncology, and AstraZeneca. LE has received consulting fees from Merck Sharp and Dohme, Bristol Myers Squibb, and Amgen; has received honorariums from Merck Sharp and Dohme, Bristol Myers Squibb, and Servier; and has received support for attending meetings and/or travel from Servier and Merck Serono. HW is an employee of BeiGene and has stock or stock options in BeiGene. YX, JL, TS, and SY are employees of BeiGene and have no additional competing interests. LL is an employee of BeiGene (Beijing); has received support for attending meetings and/or travel from BeiGene (Beijing); and has stock or stock options with BeiGene (Beijing). MM has received grants from Amgen, Leap Therapeutics, Merck Serono, AstraZeneca, and Merck Sharp and Dohme; has received consulting fees from Bayer, Merck Sharp and Dohme, Merck Serono, Amgen, Taiho Pharmaceutical, Pfizer, F Hoffmann-La Roche, Lilly, Servier, BeiGene, Bristol Myers Squibb, and AstraZeneca; has received honorariums from Amgen, Genentech/F Hoffmann-La Roche, Merck Serono, Merck Sharp and Dohme Oncology, Bristol Myers Squibb, AstraZeneca/MedImmune, Servier, Pierre Fabre, and Sanofi; and has received support for attending meetings and/or travel from Amgen, Merck Serono, F Hoffmann-La Roche, Bayer, ASCO, German Cancer Society, Merck Sharp and Dohme, ESMO, and BeiGene. R-HX has received consulting fees from Hutchison, Hengrui, Junshi, Qilu, CPPC, F Hoffmann-La Roche, Merck Serono; and has participated on data safety monitoring boards or advisory boards for Astellas, Merck Sharp and Dohme, AstraZeneca, Junshi, Hengrui, BeiGene, Innovent, CPPC, and Keymed Biosience. All other authors declare no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

Ethical approval: This study was approved by the relevant institutional review board or independent ethics committee for each study site (see supplementary appendix). The study was conducted

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.

in accordance with the International Conference on Harmonisation E6 guideline for Good Clinical Practice and the principles of the Declaration of Helsinki, and all applicable local laws and regulations. All patients were required to sign written informed consent before participation in the study.

Data sharing: On request, and subject to specific criteria, conditions, and exceptions, BeiGene will provide access to individual deidentified participant data from BeiGene sponsored global interventional clinical studies conducted for medicines for indications that have been approved, or in programmes that have been terminated. BeiGene will also consider requests for the protocol, data dictionary, and statistical analysis plan. Data requests can be submitted to datadisclosure@beigene.com.

Transparency: The senior author (R-HX) affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

Dissemination to participants and related patient and public communities: Results were communicated to patients who expressed an interest during clinic visits. Outcomes will be disseminated through press releases, academic conferences, and social media.

Provenance and peer review: Not commissioned; externally peer reviewed.

This is an Open Access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited and the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.

1 Sung H, Ferlay J, Siegel RL, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2021;71:209-49. doi:10.3322/caac.21660
2 GLOBOCAN. Population fact sheet: Asia. World Health Organization. 2020. https://gco.iarc.fr/today/data/factsheets/populations/935-asia-fact-sheets.pdf (accessed 27 June 2023).
3 GLOBOCAN. Population fact sheet: North America. World Health Organization. 2020. https://gco.iarc.fr/today/data/factsheets/populations/905-northern-america-fact-sheets.pdf (accessed 27 June 2023).
4 GLOBOCAN. Population fact sheet: Europe. World Health Organization. 2020. https://gco.iarc.fr/today/data/factsheets/populations/908-europe-fact-sheets.pdf (accessed 27 June 2023).
5 Cao W, Chen HD, Yu YW, Li N, Chen WQ. Changing profiles of cancer burden worldwide and in China: a secondary analysis of the global cancer statistics 2020. *Chin Med J (Engl)* 2021;134:783-91. doi:10.1097/CM9.0000000000001474
6 GLOBOCAN. Population fact sheet: Japan. World Health Organization. 2020. https://gco.iarc.fr/today/data/factsheets/populations/392-japan-fact-sheets.pdf (accessed 27 June 2023).
7 GLOBOCAN. Population fact sheet: Republic of Korea. World Health Organization. 2020. https://gco.iarc.fr/today/data/factsheets/populations/410-korea-republic-of-fact-sheets.pdf (accessed 28 June 2023).
8 Lordick F, Carneiro F, Cascinu S, et al, ESMO Guidelines Committee. Electronic address: clinicalguidelines@esmo.org. Gastric cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Ann Oncol* 2022;33:1005-20. doi:10.1016/j.annonc.2022.07.004
9 Obermannová R, Alsina M, Cervantes A, et al, ESMO Guidelines Committee. Electronic address: clinicalguidelines@esmo.org. Oesophageal cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Ann Oncol* 2022;33:992-1004. doi:10.1016/j.annonc.2022.07.003
10 Cheng J, Cai M, Shuai X, Gao J, Wang G, Tao K. First-line systemic therapy for advanced gastric cancer: a systematic review and network meta-analysis. *Ther Adv Med Oncol* 2019;11:1758835919877726. doi:10.1177/1758835919877726
11 Catenacci DV, Chao J, Muro K, et al. Toward a treatment sequencing strategy: a systematic review of treatment regimens in advanced gastric cancer/gastroesophageal junction adenocarcinoma. *Oncologist* 2021;26:e1704-29. doi:10.1002/onco.13907
12 Shen L, Bai Y, Lin X, et al. First-line (1L) nivolumab (NIVO) plus chemotherapy (chemo) vs chemo in patients (pts) with advanced gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma (GC/GEJC/EAC): CheckMate 649 Chinese subgroup analysis with 3-year follow-up. *J Clin Oncol* 2023;41:353. doi:10.1200/JCO.2023.41.4_suppl.353.
13 Janjigian YY, Shitara K, Moehler M, et al. First-line nivolumab plus chemotherapy versus chemotherapy alone for advanced gastric, gastro-oesophageal junction, and oesophageal adenocarcinoma (CheckMate 649): a randomised, open-label, phase 3 trial. *Lancet* 2021;398:27-40. doi:10.1016/S0140-6736(21)00797-2
14 Kang YK, Chen LT, Ryu MH, et al. Nivolumab plus chemotherapy versus placebo plus chemotherapy in patients with HER2-negative, untreated, unresectable advanced or recurrent gastric or gastro-oesophageal junction cancer (ATTRACTION-4): a randomised, multicentre, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol* 2022;23:234-47. doi:10.1016/S1470-2045(21)00692-6
15 Shitara K, Van Cutsem E, Bang YJ, et al. Efficacy and safety of pembrolizumab or pembrolizumab plus chemotherapy vs chemotherapy alone for patients with first-line, advanced gastric cancer: the KEYNOTE-062 phase 3 randomized clinical trial. *JAMA Oncol* 2020;6:1571-80. doi:10.1001/jamaoncol.2020.3370
16 Rha SY, Oh D-Y, Yañez P, et al, KEYNOTE-859 investigators. Pembrolizumab plus chemotherapy versus placebo plus chemotherapy for HER2-negative advanced gastric cancer (KEYNOTE-859): a multicentre, randomised, double-blind, phase 3 trial. *Lancet Oncol* 2023;24:1181-95. doi:10.1016/S1470-2045(23)00515-6
17 Xu J, Jiang H, Pan Y, et al, ORIENT-16 Investigators. Sintilimab plus chemotherapy for unresectable gastric or gastroesophageal junction cancer: the ORIENT-16 randomized clinical trial. *JAMA* 2023;330:2064-74. doi:10.1001/jama.2023.19918
18 Hong Y, Feng Y, Sun H, et al. Tislelizumab uniquely binds to the CC' loop of PD-1 with slow-dissociated rate and complete PD-L1 blockage. *FEBS Open Bio* 2021;11:782-92. doi:10.1002/2211-5463.13102
19 Feng Y, Hong Y, Sun H, et al. Abstract 2383: The molecular binding mechanism of tislelizumab, an investigational anti-PD-1 antibody, is differentiated from pembrolizumab and nivolumab. *Cancer Res* 2019;79:2383. doi:10.1158/1538-7445.AM2019-2383.
20 Zhang T, Song X, Xu L, et al. The binding of an anti-PD-1 antibody to FcγRI has a profound impact on its biological functions. *Cancer Immunol Immunother* 2018;67:1079-90. doi:10.1007/s00262-018-2160-x
21 Xu J, Bai Y, Xu N, et al. Tislelizumab plus chemotherapy as first-line treatment for advanced esophageal squamous cell carcinoma and gastric/gastroesophageal junction adenocarcinoma. *Clin Cancer Res* 2020;26:4542-50. doi:10.1158/1078-0432.CCR-19-3561
22 Liu C, Fang F, Kong Y, ElGabry EA. Tumor Area Positivity (TAP) score of programmed death-ligand 1 (PD-L1): a novel visual estimation method for combined tumor cell and immune cell scoring. *Diagn Pathol* 2023;18:48. doi:10.1186/s13000-023-01318-8
23 Ellis LM, Bernstein DS, Voest EE, et al. American Society of Clinical Oncology perspective: Raising the bar for clinical trials by defining clinically meaningful outcomes. *J Clin Oncol* 2014;32:1277-80.
24 US Prescribing information. OPDIVO (nivolumab) injection, for intravenous use. 2023. https://packageinserts.bms.com/pi/pi_opdivo.pdf (accessed 8 August 2023).

Supplementary material: Additional information, tables S1-S12, figures S1-S5, study protocol, and statistical analysis plan

BMJ: first published as 10.1136/bmj-2023-078876 on 28 May 2024. Downloaded from http://www.bmj.com/ on 23 August 2024 by guest. Protected by copyright.